

18 MILITARY INNOVATIONS THAT WILL RULE THE WORLD



01

AI SURVEILLANCE

Militaries around the world capture heaps of data from all kinds of surveillance devices, including CCTV cameras, satellites, drones and even personal computing devices. The challenge is to churn this vast amount of data into useful strategic information. Militaries around the world are using AI here. One of the most important uses is object recognition that can track build up of facilities, troop movements and even an individual moving across a city.

02

ANTI DRONE WEAPONS

With the advent of drones for military purposes, there is a requirement for weapons to take down drones. These can include towers that fire lasers at drones, jamming guns that disrupt the connections, and even other drones that are sent up specifically to take down enemy drones. Perhaps the most innovative approach was by the police in the Netherlands, who trained eagles to hunt down drones.

03

AUTONOMOUS CONVOYS

One of the biggest benefits to logistics in the future is through autonomous convoys for moving goods. The vehicles in such convoys save fuel by moving in tight formation at steady speeds. Militaries are exploring how to apply these benefits for convoys. The convoys could

Military research around the world is focused on developing autonomous vehicles, high energy weapons and defensive techniques all designed to save more human lives

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send in supplies into combat zones. Sending convoys through routes that are particularly susceptible to enemy attacks, can help reduce casualties.

04

AUVS AND UUVS

An emerging class of vehicles are called automatic underwater vehicles or unmanned underwater vehicles. These are stealth submarines that provide long term surveillance in the seas. One of the design constraints of such vehicles is that they can interface with satellites only when they surface, so they cannot be remote controlled very easily.

05

BLIMPS

Airships are making a comeback, because of the benefits of autonomy and how cheap they are. Blimps can be easily deployed in remote areas, and can provide surveillance for months. Such setups could be used along borders to prevent infiltration and monitor any activities.

06

CLOAKING DEVICES

In the future, we will see an increasing number of cloaking devices that make vehicles

hard to detect. There are already tiles that heat up to change how a vehicle appears in infrared light, so a tank can disguise itself as a car when seen through night vision equipment. Such technology can be expected to spread across the electromagnetic spectrum. There are also techniques to reduce the visibility of combat vehicles in radar.

07

COLOUR IR

Colour Infrared film was actually developed in World War II, to help identify camouflage military equipment, that stood out as cyan patches among vegetation that appears red in IR. The process involves taking signals from the IR spectrum, and shifting them so that they approximate the colours as seen by the human eye. The result is stunning colour images even in the night, something that can find its way to IR devices deployed to soldiers.

08

CYBORG ANIMALS

Militaries around the world have tried to rope in the animal kingdom for wars, and this reached new heights with the advent of technology. While China has